

NSF CISE CORE (IIS) Budget

CopernicusAI Knowledge Engine: Assisted Research via AI Briefings, Multi-Modal Metadata, and Process Visualization

Total Budget Request: \$691,184 **Project Period:** 36 months (3 years) **Principal Investigator:** Gary Welz

Budget Summary

Category	Year 1	Year 2	Year 3	Total	Percentage
A. Senior Personnel	\$60,000	\$60,000	\$60,000	\$180,000	35%
B. Other Personnel	\$26,000	\$26,000	\$26,000	\$78,000	15%
C. Fringe Benefits	\$17,343	\$17,343	\$17,343	\$52,029	10%
D. Equipment	\$8,000	\$2,000	\$2,000	\$12,000	2%
E. Travel	\$4,000	\$6,000	\$6,000	\$16,000	3%
F. Participant Support	\$0	\$0	\$0	\$0	0%
G. Other Direct Costs	\$37,000	\$40,000	\$38,000	\$115,000	22%
Total Direct Costs	\$152,343	\$151,343	\$149,343	\$453,029	66%
H. Indirect Costs	\$77,945	\$80,645	\$79,565	\$238,155	34%
TOTAL	\$230,288	\$231,988	\$228,908	\$691,184	100%

Note: Fringe benefits use CUNY Graduate Center rates: PI summer salary 25.05%; other personnel (part-time) 8% + 0.895% MTA. Indirect cost rate: 54% (research) applied to modified total direct costs (MTDC). MTDC excludes equipment. Materials & Supplies total \$15,000.

Detailed Budget Breakdown

A. Senior Personnel (\$180,000 total)

Principal Investigator (Gary Welz) - Year 1: 6 months @ \$10,000/month = \$60,000 - Year 2: 6 months @ \$10,000/month = \$60,000 - Year 3: 6 months @ \$10,000/month = \$60,000 - **Total: \$180,000**

Justification: The PI will lead all research activities including interface design, metadata schema development, reliability mechanism implementation, and evaluation studies. Six months per year effort reflects the technical and research-intensive nature of the work, while allowing time for other professional activities.

B. Other Personnel (\$78,000 total)

Research Assistant / Software Developer - Year 1: 4 months @ \$5,000/month = \$20,000 - Year 2: 4 months @ \$5,000/month = \$20,000 - Year 3: 4 months @ \$5,000/month = \$20,000 - **Subtotal: \$60,000**

Student User Testers and Collaborators (Alpha/Beta Testing) - Year 1 (Alpha Testing): - 2 Graduate Students @ \$15/hour, 10 hours/week, 20 weeks = \$6,000 - **Year 1 Subtotal: \$6,000**

- **Year 2 (Expanded Beta Testing):**
 - 2 Graduate Students @ \$15/hour, 10 hours/week, 20 weeks = \$6,000
 - **Year 2 Subtotal: \$6,000**
- **Year 3 (Full Beta and Validation):**
 - 2 Graduate Students @ \$15/hour, 10 hours/week, 20 weeks = \$6,000
 - **Year 3 Subtotal: \$6,000**
- **Student Testing Subtotal: \$18,000**

Note: Budget revised to focus on graduate students only. Undergraduate students removed to simplify personnel management and align with Research.gov budget form requirements.

Total Other Personnel: \$78,000

Justification: - **Research Assistant:** Additional development capacity needed for implementation of metadata pipelines, cross-modal linking systems, and evaluation infrastructure. Effort increases in Years 2-3 as system complexity grows and user studies begin.

- **Student User Testers:** Critical component for usability testing, collaborative validation, and grounding the system in real-world use. Students will:
 - Use the platform to create research briefings and podcasts
 - Provide feedback on usability and functionality
 - Participate in collaborative validation of AI outputs
 - Test iterative refinement workflows
 - Generate real-world usage data for evaluation
 - Contribute to collective knowledge building through shared artifacts

This approach not only tests system usability but also establishes collaborative validation mechanisms where multiple users verify and refine AI-generated content, creating a collective intelligence layer that improves system reliability and trustworthiness. The phased approach (alpha → expanded beta → full beta) allows for iterative improvements based on user feedback.

C. Fringe Benefits (\$52,029 total)

Rates per CUNY Graduate Center: summer salary 25.05%; part-time personnel 8% + 0.895% MTA.

Principal Investigator (summer salary): - 25.05% x \$60,000/year = \$15,030/year - **Total: \$45,090**

Research Assistant (part-time, 4 months/year): - (8% + 0.895% MTA) x \$20,000/year = \$1,779/year - **Total: \$5,337**

Student User Testers (part-time): - (8% + 0.895% MTA) x \$6,000/year = \$534/year - **Total: \$1,602**

Fringe total: \$52,029 (\$17,343 per year)

D. Equipment (\$12,000 total)

Year 1: - High-performance development laptop, development tools, and software: \$5,000 - Monitoring and analytics tools, and cloud service credits/reserves: \$3,000 - **Year 1 Total: \$8,000**

Note: Equipment description split into two line items to accommodate detailed descriptions within Research.gov form field limitations.

Year 2: - Equipment 1: \$0 (no new dev laptop) - Software renewals, cloud service credits/reserves: \$2,000 (Equipment 2) - **Year 2 Total: \$2,000**

Year 3: - Software renewals, cloud service credits/reserves: \$2,000 - **Year 3 Total: \$2,000**

Justification: Enhanced development capabilities for complex system integration work. Cloud service credits provide buffer for unexpected usage during intensive development and testing phases.

E. Travel (\$16,000 total)

Year 1: - NSF PI Meeting / CISE Conference: \$2,000 - Relevant AI/Research Tools Conference: \$2,000 - **Year 1 Total: \$4,000**

Year 2: - NSF PI Meeting / CISE Conference: \$2,000 - AI/Research Tools Conference (presentation): \$2,000 - User Study Site Visits: \$2,000 - **Year 2 Total: \$6,000**

Year 3: - NSF PI Meeting / CISE Conference: \$2,000 - Final Results Presentation Conference: \$2,000 - User Study Site Visits: \$2,000 - **Year 3 Total: \$6,000**

Justification: Travel to NSF/CISE meetings, relevant conferences for staying current with developments and presenting research results, and site visits for user studies and collaboration.

F. Participant Support (\$0)

Not applicable for this project.

G. Other Direct Costs (\$115,000 total)

Cloud Infrastructure & Services: \$78,000 total (Year 1: \$25,000; Year 2: \$28,000; Year 3: \$25,000)

Consultants/Subcontracts: \$15,000 total (Year 1: \$5,000; Year 2: \$5,000; Year 3: \$5,000)

Materials & Supplies: \$15,000 total (Year 1: \$5,000; Year 2: \$5,000; Year 3: \$5,000)

Publication & Documentation: \$4,000 total (Year 1: \$1,000; Year 2: \$1,000; Year 3: \$2,000)

Other: \$3,000 total (Year 1: \$1,000; Year 2: \$1,000; Year 3: \$1,000)

Year 1 Other Direct: \$37,000 | Year 2: \$40,000 | Year 3: \$38,000 | Total: \$115,000

Justification: Cloud infrastructure supports production-scale development and multi-model LLM services. Consultants provide metadata design, quality assurance, and evaluation expertise. Materials and publication costs support documentation and dissemination.

H. Indirect Costs (\$238,155 total)

Indirect Cost Rate: 54% (CUNY Graduate Center negotiated rate for research) applied to modified total direct costs (MTDC). MTDC excludes equipment.

- Year 1: $54\% \times (\$152,343 - \$8,000) = 54\% \times \$144,343 = \mathbf{\$77,945}$
- Year 2: $54\% \times (\$151,343 - \$2,000) = 54\% \times \$149,343 = \mathbf{\$80,645}$
- Year 3: $54\% \times (\$149,343 - \$2,000) = 54\% \times \$147,343 = \mathbf{\$79,565}$
- **Total: \$238,155**

Justification: CUNY Graduate Center negotiated indirect cost rate for research (54%).

Budget Justification Summary

This budget supports a 3-year research project to develop and evaluate an AI-enabled assisted research platform. The budget allocates:

- **35% to Senior Personnel:** PI effort (6 months/year) for research leadership and technical development
- **15% to Other Personnel:** Research assistant/developer support plus student user testers for alpha/beta testing and collaborative validation
- **10% to Fringe Benefits:** CUNY GC rates (PI summer 25.05%; part-time personnel 8% + 0.895% MTA)
- **22% to Other Direct Costs:** Cloud infrastructure, consultants, materials (\$15,000), publication, other
- **34% to Indirect Costs:** CUNY Graduate Center research rate (54% of MTDC)

Key Cost Drivers: 1. **Personnel (67% combined):** Research-intensive work requires substantial PI and support personnel effort, plus student user testing for usability evaluation and collaborative validation 2. **Cloud Infrastructure (17% of total):** Production GCP services and multi-model LLM API costs are essential for the research 3. **Travel (4%):** Necessary for NSF meetings, conferences, and user study site visits

Student User Testing Component: The student user testing allocation (\$80,160 over 3 years) supports a critical research component: collaborative validation and real-world usability testing. Students will use the platform to create research briefings, test workflows, and participate in collective verification of AI outputs. This approach grounds the research in actual use cases while establishing mechanisms for collaborative intelligence that improve system reliability.

Year-by-Year Variations: - **Year 1:** Higher equipment costs for initial setup - **Year 2:** Increased personnel and cloud costs as user studies begin - **Year 3:** Focus on evaluation and documentation, slightly lower cloud costs

This budget is consistent with typical NSF CISE Core awards and supports the research objectives outlined in the proposal. The allocation reflects the technical nature of the work, the need for production-scale infrastructure, and the importance of evaluation and dissemination activities.

Cost Sharing

No cost sharing is proposed for this project.

Budget Prepared By: Gary Welz, Principal Investigator

Total Direct Costs Requested: \$453,029

Total Indirect Costs: \$238,155 (54% of MTDC; CUNY GC research rate)

Total Request: \$691,184

Note: Fringe and indirect rates per CUNY Graduate Center Sponsored Research. Materials & Supplies total \$15,000.